Converted vs. Unconverted Photons: Shower Shape, Isolation, and Energy Response in photon10 MC

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Abstract

We study the impact of photon conversions (pre-EMCal e^+e^- pair production in detector material) on reconstruction observables used by the PPG12 isolated-photon cross-section measurement in p+p at $\sqrt{s} = 200$ GeV. Using the `photon10` single-particle Monte Carlo (~ 10 M events, truth $p_T \in [14, 30]$ GeV), we find that **17.3%** of fiducial truth photons ($|\eta| < 0.7$, $14 \leq p_T < 30$ GeV) convert before the EMCal. Converted photons exhibit (i) a mean cluster-energy response shift of $\Delta\langle R_E\rangle \simeq -0.076$ (-8%) driven by an asymmetric low-R tail, (ii) a drop of -14.6 percentage points in the parametric reco-isolation pass fraction (rising to -22.8 pp at 8–10 GeV), dominated by additional EMCal energy inside the isolation cone, and (iii) a -16% shift in the mean shower-shape BDT score, with w_ϕ^{CoG} broadening by $+91\%$ due to the solenoidal-field opening of the e^+e^- pair. These four features are mutually consistent with a single physical picture: the ~ 1.4 T sPHENIX solenoid splits the e^+e^- pair in azimuth, one arm leaks outside the cluster footprint (reducing the energy response and polluting the isolation cone), and the resulting wider, two-lobed ϕ-profile degrades the ID BDT. The effect enters the cross-section measurement as a joint ID+iso efficiency systematic whose size is set by the accuracy of the simulated material budget.

1 Introduction and Motivation

Photon conversion — the reaction $\gamma \to e^+e^-$ induced by the Coulomb field of a nucleus — occurs with probability $P \simeq 1 - \exp\left[-\frac{7}{9}\, X/X_0\right]$ as a photon traverses a thickness X/X_0 of material. In sPHENIX the tracker (MVTX + INTT + TPC) together with its support structures and cables precedes the barrel EMCal. With the nominally quoted $\sim$ 10–20% X_0 of material, a $\sim$ 10–15% conversion probability is expected in the central barrel. Once a photon converts, the two leptons are bent in opposite azimuthal directions by the ~ 1.4 T solenoid, land in the EMCal as two partially overlapping electromagnetic showers, and are reconstructed by the standard clustering as a single (wider) cluster whose properties differ from those of a pristine photon.

These distortions matter for PPG12 because the tight-selected isolated photon signal efficiency is the product of a shower-shape identification efficiency (via the E_T-dependent BDT cut) and an isolation efficiency (via the parametric isolation cone cut). Both steps are sensitive to the converted fraction and its pre-reconstruction kinematics. Any mismatch in the simulated material budget between Monte Carlo and data therefore propagates directly into the cross-section. Quantifying the conversion fingerprint on each observable — and its p_T and η dependence — is a prerequisite for assigning a material-budget systematic uncertainty.

Truth conversion information is carried in the slimtree through the `particle_converted` flag written by `anatreemaker/source/CaloAna24.cc`. The flag is built by traversing the G4 shower particle list of each truth photon (`CaloAna24.cc` lines $\sim$ 807–843): any secondary that (a) originates at a G4 vertex with $r_\perp < 93$ cm (i.e. upstream of the EMCal front face) and (b) carries more than 40% of the photon momentum, tags the photon. If at least one such secondary is an $e^\pm$ ($|\mathrm{pid}| = 11$), the photon is labelled `converted`; otherwise it is labelled `bad`. The flag is written at `CaloAna24.cc`

lines $\sim$ 953–970 using:

$$\texttt{particle_converted} = \begin{cases} 0 & \text{no upstream secondary above threshold (unconverted)} \\ 1 & e^+e^- \text{ daughter above threshold (converted)} \\ 2 & \text{non-}e^+e^- \text{ secondary above threshold (bad)} \end{cases} \tag{1}$$

In what follows we treat the small "bad" population (0.01–0.02% of truth photons, < 300 matched in 10 M events) as physically negligible and focus on the unconverted–converted comparison.

2 Data Sample and Method

Sample. The study is based on the single-photon Monte Carlo file `photon10_with_bdt_vtx_noreweight_singl` (one fired photon per event, truth $p_T \in [14, 30]$ GeV at generation, $|\eta| < 1.5$). Four independent analyses read the tree `slimtree`:

- Fractions, shower shape, and energy response use the full 9,998,541 events.
- Isolation uses a 2,000,000-event subsample (runtime-based).

Truth selection (fiducial). `particle_pid` $== 22$, $|$`particle_Eta`$| < 0.7$, `particle_truth_iso_03` $<$ 4 GeV, and (where applicable) $8 \leq$ `particle_Pt` ≤ 36 GeV or $14 \leq$ `particle_Pt` < 30 GeV as indicated.

Cluster–truth matching. For each truth photon the highest-E_T cluster in `CLUSTERINFO_CEMC` whose `cluster_truthtrkID` equals the truth `particle_trkid` is selected. A residual cluster E_T threshold ($>$ 3–5 GeV, depending on the analysis) is applied. pT binning follows the PPG12 analysis edges `ptRanges` $= \{8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 32, 36\}$ (12 bins) with the matched cluster E_T as the abscissa.

Caveat on isolation. The input file *does not* contain the topo-cluster isolation branches `cluster_iso_topo_03/` The isolation study therefore uses the calo-tower isolation `cluster_iso_03` (and R=0.4 equivalent). The nominal analysis configuration `efficiencytool/config_bdt_nom.yaml` uses topo-iso at $R = 0.4$ (`use_topo_iso: 2`). The *shape* of the converted/unconverted difference is expected to carry over, but absolute magnitudes should be re-measured on topo-iso branches once they are populated for `photon10`.

Shower shape caveat. The shower-shape BDT score is only written to the `CLUSTERINFO_CEMC_NO_SPLIT` node, whereas the other shower-shape observables use `CLUSTERINFO_CEMC`. BDT conclusions below are therefore based on the unsplit-cluster node.

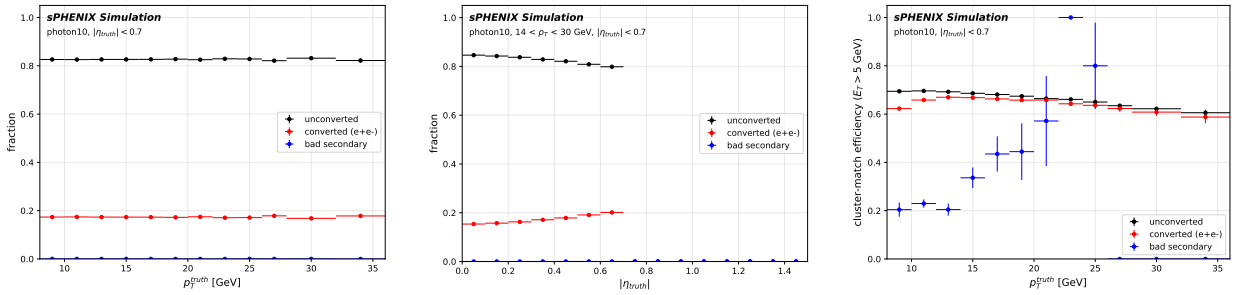
3 Conversion Fractions and Cluster Match Efficiency

In the fiducial region ($|\eta| < 0.7$, $14 \leq p_T < 30$ GeV, 855,380 truth photons), the converted fraction is $(17.317 \pm 0.041)\%$ and the "bad" fraction is a negligible $(0.024 \pm 0.002)\%$. Table 1 summarises the inclusive category counts and per-category cluster-match efficiency. The converted fraction is effectively flat in p_T (17.2–17.9% across 8–36 GeV) and rises with $|\eta|$, from $(15.4 \pm 0.3)\%$ at $|\eta| < 0.1$ to $(20.1 \pm 0.4)\%$ at $0.6 < |\eta| < 0.7$, reaching 32% at $|\eta| \simeq 1.4$–1.5 (extended window).

Table 1: Converted-photon fractions and cluster-match efficiency in the fiducial window $|\eta| < 0.7$, $14 \leq p_T < 30$ GeV. Match efficiency is defined as the fraction of truth photons (within a category) for which at least one cluster satisfies `cluster_truthtrkID` == `particle_trkid` and `cluster_Et` > 5 GeV.

Category	N (truth)	Fraction	Match efficiency
Unconverted	707,047	$(82.659 \pm 0.041)\%$	$(67.987 \pm 0.055)\%$
Converted (e^+e^-)	148,129	$(17.317 \pm 0.041)\%$	$(66.290 \pm 0.123)\%$
Bad secondary	204	$(0.024 \pm 0.002)\%$	$(39.024 \pm 3.407)\%$

The three panels of Fig. 1 show the main trends:



(a) Fractions vs. p_T ($|\eta| < 0.7$) (b) Fractions vs. $|\eta|$ ($14 \leq p_T < 30$ GeV) (c) Cluster-match efficiency vs. p_T

Figure 1: Converted-photon fractions and match efficiency in the fiducial window. The converted fraction is flat in p_T and rises with $|\eta|$ consistent with increased integrated material at forward rapidity. Converted photons have ~ 2 pp lower cluster-match efficiency than unconverted.

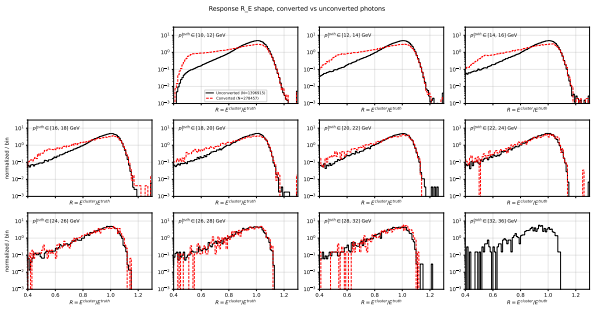
The central-barrel converted fraction of $\sim 17\%$ is consistent with a material budget near the upper end of the nominally quoted 10–20% X_0 range ($1 - \exp[-\frac{7}{9} \cdot 0.20] \simeq 15\%$). The sharp rise toward $|\eta| \sim 1.5$ is consistent with endcap-style material growth and matches expectations for the tracker support geometry.

The absolute match-efficiency value ($\sim$ 66–68% for fiducial photons) looks low at first glance because the definition *requires* `cluster_truthtrkID` to equal the *primary* photon track id. When clustering associates a post-shower secondary ($e^{\pm}$, brem γ) instead of the original photon, the match fails even though a physically correct cluster is present. The converted-category match efficiency is therefore a *lower bound*; the *relative* $\sim 2.5\%$ deficit with respect to unconverted is the physics-meaningful quantity.

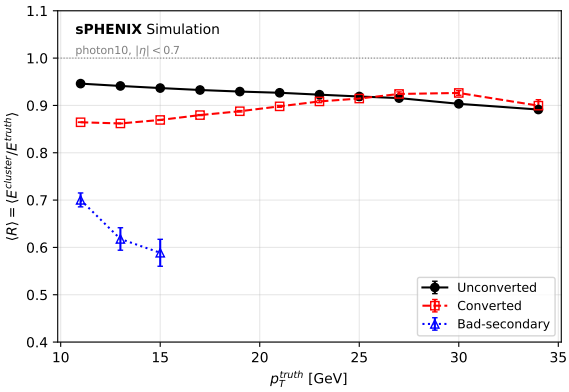
4 Energy Response and Resolution

We define the cluster energy response $R_E = E_{\text{cluster}}/E_{\gamma}^{\text{truth}}$ and compare its shape, first and second moments, and low-R tail fraction between the converted and unconverted categories. Analogous results for R_{E_T} are contained in the auxiliary grid `response_shape_ET_grid.pdf`.

4.1 Shape, mean, and resolution

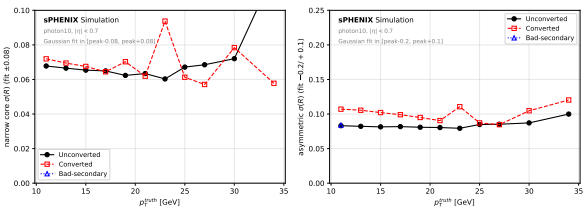


(a) Response shape R_E per p_T bin

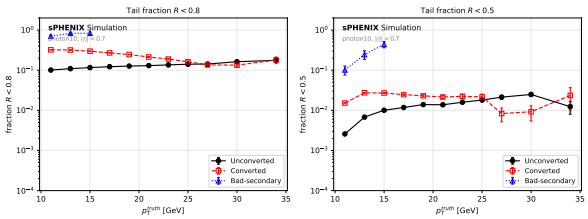


(b) Mean response $\langle R_E \rangle$ vs. p_T

Figure 2: Response distributions and means for unconverted vs. converted photons. The converted distribution shows a pronounced low-R tail at $p_T \lesssim 20$ GeV caused by one arm of the e^+e^- pair leaking outside the cluster footprint. Above $p_T \sim 25$ GeV the distributions converge and the mean response of converted photons crosses above that of unconverted (pair-collimation limit).



(a) Core and asymmetric σ vs. p_T



(b) Tail fractions $R < 0.8$ and $R < 0.5$ vs. p_T

Figure 3: Resolution and tail-fraction summary. The Gaussian core width is only $\sim 5\%$ wider for converted photons, but an asymmetric fit covering the left tail yields $\sim 27\%$ broadening. The tail population drives the mean shift.

Table 2: Inclusive ($10 \leq p_T \leq 34$ GeV) energy-response summary for converted vs. unconverted photons. Narrow-core fit: Gaussian in [peak−0.08, peak+0.08]. Asymmetric fit: [peak−0.2, peak+0.1] (captures the low-R tail).

Quantity	Unconverted	Converted	Converted − Unconverted
$\langle R_E \rangle$	0.9424	0.8666	−0.0758 (−8%)
median R_E	0.9650	0.9050	−0.060
upper width $p_{84} - p_{50}$	0.080	0.120	+50%
lower width $p_{50} - p_{16}$	0.120	0.230	+92%
narrow-core μ	0.9956	0.9994	+0.004
narrow-core σ	0.0669	0.0703	+5%
asymmetric σ	0.0826	0.1051	+27%
fraction $R < 0.8$	0.107	0.311	×2.9
fraction $R < 0.5$	0.0053	0.0201	×3.8

The mean shift of $\Delta\langle R_E \rangle = -0.076$ is not a shift of the core: the narrow-core μ is actually slightly higher for converted photons (the surviving core reconstructs at unity), and the narrow-core σ

broadens by only $\sim 5\%$. The apparent -8% mean shift and the $+27\%$ asymmetric σ are driven entirely by an asymmetric low-R tail where one of the e^+e^- arms deposits less than 90% of its expected energy in the cluster footprint. The tail fraction below $R = 0.8$ is $\sim 3\times$ larger for converted photons and below $R = 0.5$ is $\sim 4\times$ larger.

4.2 p_T trend and crossover

Per-p_T shifts (from `response_findings.md` Sec. 4): $\Delta\langle R_E\rangle \simeq -0.082$ at 10–12 GeV, shrinking to -0.042 at 18–20 GeV, -0.014 at 22–24 GeV, and *crossing sign* at $\sim$ 26–27 GeV. Above $p_T \simeq 25$ GeV the mean response of converted photons *exceeds* that of unconverted (by up to $+0.023$ at 28–32 GeV). Physically, the e^+e^- opening angle after magnetic-field bending decreases as $\sim 1/p_T$, so the pair becomes collimated enough to be contained in a single cluster — exactly where the *unconverted* response continues to degrade slowly with p_T from shower leakage into the HCal. The same crossover appears in the tail-0.8 curves (converted tail decreases from 0.32 to 0.13 as p_T rises; unconverted tail rises mildly from 0.10 to 0.17), confirming the effect.

4.3 Angular deltas

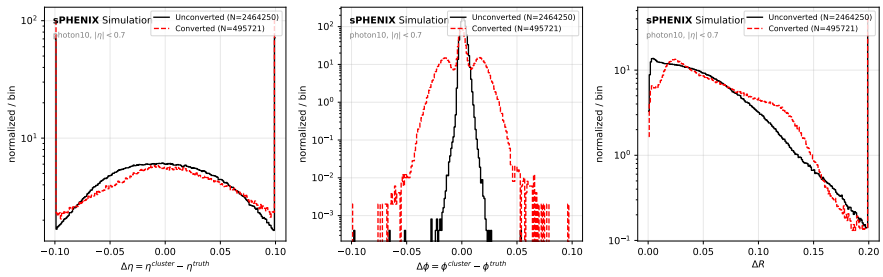


Figure 4: Cluster-minus-truth angular deltas $\Delta\eta$, $\Delta\phi$ and ΔR. The $\Delta\phi$ distribution for converted photons is strikingly bimodal with lobes at $\Delta\phi \sim \pm 0.01$ rad and a dip at zero — the signature of magnetic-field bending of the e^+e^- pair, with the cluster centroid shifting toward the higher-energy lepton. $\Delta\eta$ shows no shift or broadening, consistent with a purely azimuthal magnetic field. ΔR has a broader tail for converted photons (up to 0.15 rad) that correlates with the $\Delta\phi$ lobes.

The $\Delta\phi$ bimodality is a purely kinematic consequence of the e^+e^- pair carrying opposite charges: the solenoidal field bends the leptons in opposite senses with equal probability, so the mean $\Delta\phi$ is zero while the RMS is roughly an order of magnitude larger than that of unconverted photons. This feature is used below (Sec. 6) to interpret the w_ϕ^{CoG} distributions.

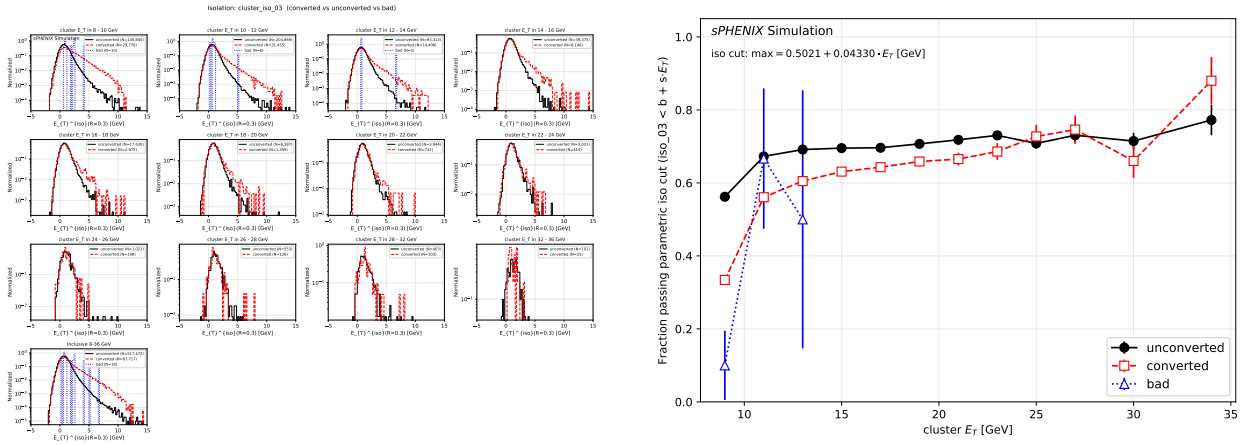
5 Isolation Energy

The PPG12 nominal reconstruction isolation cut is the parametric form

$$E_T^{\mathrm{iso}}\ \max = 0.502095 + 0.0433036 \cdot E_T, \tag{2}$$

given in GeV with E_T also in GeV. In the nominal analysis the input variable is the topo-cluster isolation `cluster_iso_topo_04` (`use_topo_iso: 2` in `config_bdt_nom.yaml`). Because the input MC file lacks that branch, we apply the same cut to `cluster_iso_03` (calo-tower iso, own-cluster towers removed). Absolute numbers below are tied to this substitution; the qualitative conclusions on converted–unconverted separation are robust because both isolation definitions sum tower E_T in an $R = 0.3$ cone.

5.1 Parametric cut pass fraction



(a) `cluster_iso_03` shape per p_T bin

(b) Parametric-cut pass fraction vs. p_T

Figure 5: Isolation distributions (left) and parametric-cut pass fraction (right). The low-p_T converted population has a long tail to $E_T^{\text{iso}} > 2$ GeV that fails the parametric cut; the inclusive pass-fraction deficit is -14.6 pp for converted photons, dominated by the 8–14 GeV region.

Inclusive ($8 \leq E_T < 36$ GeV) pass fractions: `unconv` $= 0.649$, `conv` $= 0.503$, $\Delta = -0.146$ (i.e. a -14.6 **percentage-point** drop). The deficit peaks at -22.8 pp in the 8–10 GeV bin ($0.562 \rightarrow 0.334$) and shrinks to the ± 5 pp statistics-limited level above 24 GeV.

5.2 Mean isolation energy and calorimeter decomposition

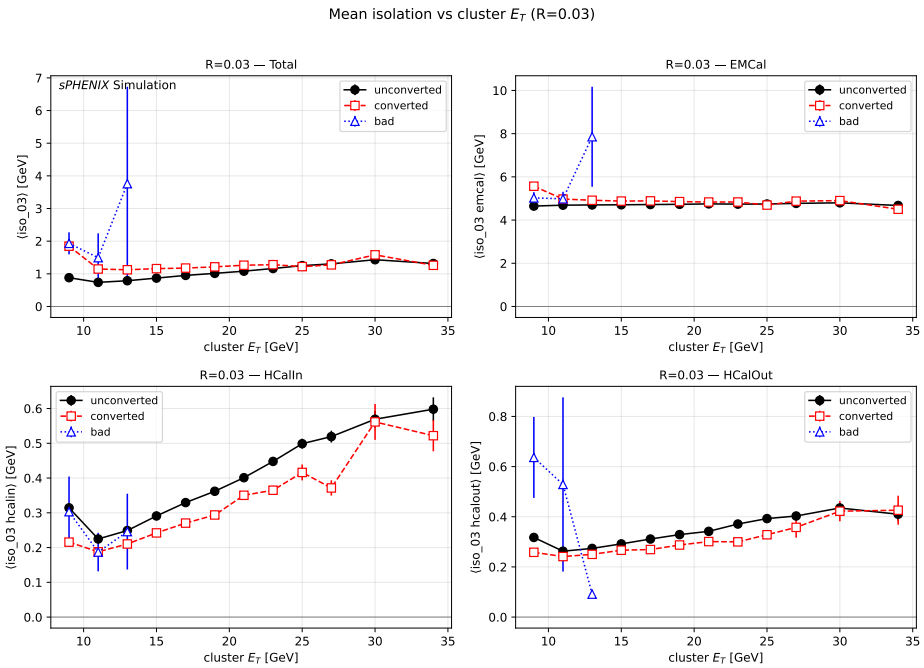


Figure 6: Mean isolation energy $\langle$`cluster_iso_03`$\rangle$ vs. cluster E_T, split into total + EMCal + inner HCal + outer HCal sub-components. The total and EMCal sub-components exhibit a clear converted/unconverted separation; both HCal sub-components show the opposite (small) trend.

Table 3: Inclusive ($8 \leq E_T < 36$ GeV) mean isolation energies for the total cone and its three sub-components (EMCal, inner HCal, outer HCal) at $R = 0.3$ and $R = 0.4$. All values in GeV. Note that `cluster_iso_03_emcal` is the full EMCal cone integral minus the bare cluster E_T — it is a diagnostic, not a decomposition of the total.

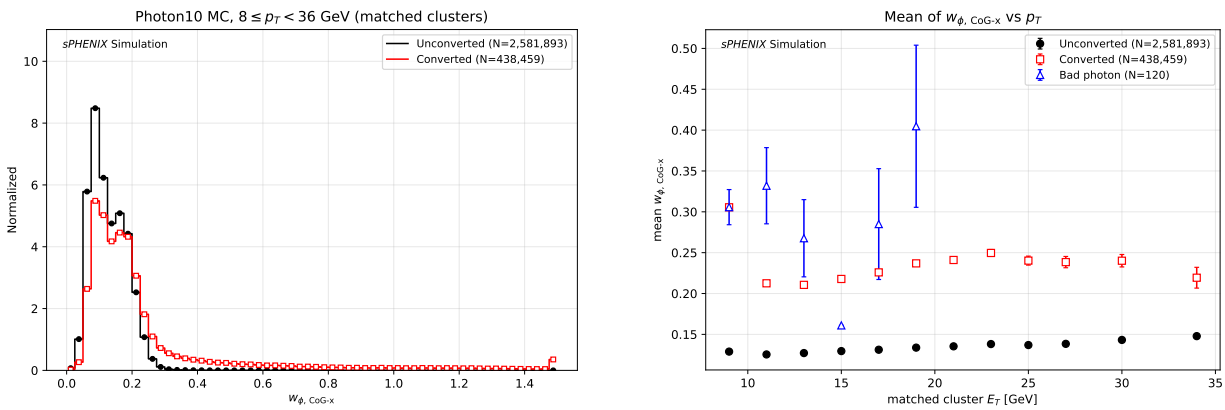
Variable	Unconverted	Converted	Converted − Unconverted
`cluster_iso_03` (total)	0.816	1.386	+0.570
`cluster_iso_03_emcal`	4.683	5.151	+0.468
`cluster_iso_03_hcalin`	0.269	0.213	−0.056
`cluster_iso_03_hcalout`	0.287	0.253	−0.033
`cluster_iso_04` (total)	0.957	1.531	+0.574
`cluster_iso_04_emcal`	8.166	8.484	+0.318
`cluster_iso_04_hcalin`	0.297	0.241	−0.056
`cluster_iso_04_hcalout`	0.434	0.398	−0.036

The +0.57 GeV mean shift at $R = 0.3$ is driven by the EMCal sub-component (+0.47 GeV) while both HCal sub-components move in the opposite direction (−0.05 and −0.03 GeV). The physical picture is exactly the converse of what is "leaking out" of the cluster in Sec. 4: the e^+e^- arm that curls outside the cluster footprint still deposits its shower *inside* the $R = 0.3$ isolation cone, inflating `iso_03_emcal`. Meanwhile, because the converted EMCal shower is more compactly absorbed (two overlapping e^+e^- showers), *less* energy leaks to the HCal. The same physics explains why the effect is largest at low E_T (where the bent-pair opening is largest relative to a fixed $R = 0.3$ cone) and fades at high E_T where the pair is collimated enough to fall inside the cluster.

6 Shower Shape and ID-Variable Separation

The shower-shape BDT uses the 25 features specified in `FunWithxgboost/config.yaml`. Here we examine the response of the most informative features (w_ϕ^{CoG}, w_η^{CoG}, e_{32}/e_{35}, BDT score, CNN prob) to the converted/unconverted split.

6.1 Headline: w_ϕ^{CoG} broadens by +91%



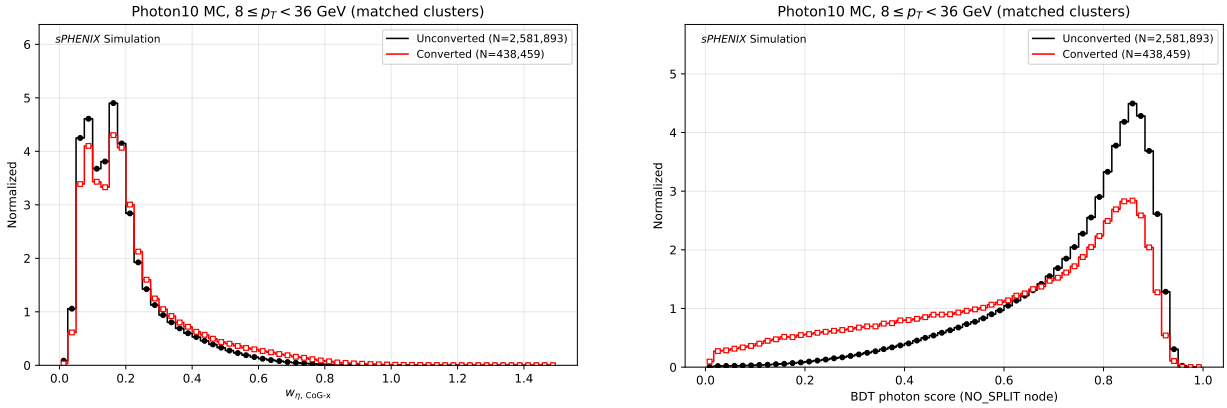
(a) Inclusive w_ϕ^{CoG} ($8 \leq E_T < 36$ GeV)

(b) Mean w_ϕ^{CoG} vs. p_T

Figure 7: w_ϕ^{CoG} (phi cluster width) inclusive distribution and p_T-binned mean. Converted photons show a clear bimodal structure — one population overlapping the unconverted core (pair unopened) and one at high w_ϕ^{CoG} (pair opened). The resulting inclusive mean shift is +0.116 (+91% relative, $2.15\,\sigma_{\mathrm{unconv}}$).

The bimodality in Fig. 7 matches exactly the bimodal $\Delta\phi$ seen in Fig. 4. Both arise from the same physics: the ~ 1.4 T sPHENIX solenoid opens the e^+e^- pair in ϕ but not in η. The resulting two-lobe cluster is wider in the bending plane, which is detected by w_ϕ^{CoG}. The PPG12 parametric tight cut (`tight_wphi_cogx_max`, implemented in `FunWithxgboost/BDTinput.C`) therefore preferentially rejects opened-pair converted photons.

6.2 Other features



(a) w_η^{CoG} inclusive

(b) Shower-shape BDT score inclusive

Figure 8: Eta cluster width (w_η^{CoG}) and shower-shape BDT score (from `CLUSTERINFO_CEMC_NO_SPLIT`). The BDT distribution for converted photons has a clear low-score tail ($\sim$ 0.3–0.5) that is essentially absent for unconverted photons.

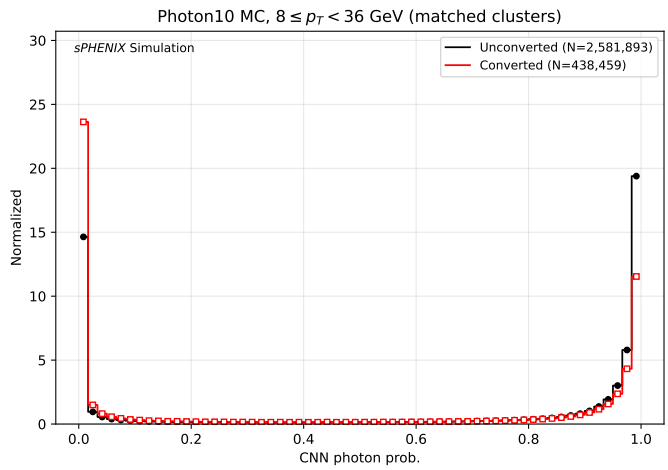


Figure 9: Inclusive CNN photon probability `CNN_prob`. Converted photons are pushed strongly toward zero (mean $0.638 \to 0.464$, -27.3% relative). The CNN is not used in the nominal selection but is by far the most discriminating single variable for converted/unconverted topology; it is shown here as a cross-check.

Table 4: Inclusive ($8 \leq E_T < 36$ GeV) means of selected shower-shape observables and signed shift (converted − unconverted). The last column is the absolute shift expressed in units of σ_{unconv}, ranking separation power.

Variable	$\langle\rangle$ unconv	$\langle\rangle$ conv	Δ (rel.)	$\lvert\Delta\rvert/\sigma_{\text{u}}$
w_η^{CoG}	0.191	0.225	+0.034 (+17.8%)	0.27
w_ϕ^{CoG}	0.128	0.244	+0.116 (+91.2%)	2.15
e_{11}/e_{33}	0.649	0.597	−0.052 (−8.0%)	0.31
e_{32}/e_{35}	0.980	0.946	−0.034 (−3.4%)	2.33
ET1 (leading)	0.939	0.911	−0.028 (−3.0%)	0.55
ET3	0.661	0.586	−0.075 (−11.3%)	0.31
CNN prob	0.638	0.464	−0.174 (−27.3%)	0.40
`cluster_prob`	0.565	0.415	−0.149 (−26.5%)	0.48
`merged_prob`	0.804	0.673	−0.131 (−16.3%)	0.55
BDT score (no-split)	0.736	0.616	−0.121 (−16.4%)	0.74

The top two discriminators (e_{32}/e_{35} at $2.33\,\sigma$ and w_ϕ^{CoG} at $2.15\,\sigma$) both test for shower broadening in the bending plane: e_{32}/e_{35} is the ratio of the central 3×2 to 3×5 tower energy, which decreases sharply when the e^+e^- pair deposits energy in the outer columns. The BDT score, by contrast, has a separation of only $0.74\,\sigma$ on the mean but shows a well-separated *low-score tail* for converted photons (Fig. 8), which drives its contribution to the tight-selection efficiency.

6.3 BDT degradation vs. p_T

Table 5: Shower-shape BDT mean shift $\Delta\langle\text{BDT}\rangle = \langle\text{BDT}\rangle_{\text{conv}} - \langle\text{BDT}\rangle_{\text{unconv}}$ vs. bin-centre p_T. The BDT degradation *grows* with p_T, driven by the growing low-score tail of converted photons. The nominal parametric tight cut is BDT $> 0.80 - 0.015 \cdot E_T$ (`bdt_min_intercept`, `bdt_min_slope` in `efficiencytool/config_bdt_nom.yaml`, lines 154–156).

p_T centre [GeV]	$\Delta\langle\text{BDT}\rangle$ (abs.)	rel. (%)
9	−0.135	−18.4
11	−0.109	−14.7
13	−0.106	−14.4
15	−0.116	−15.9
17	−0.131	−18.1
19	−0.151	−21.1
21	−0.162	−22.7
23	−0.178	−25.3
25	−0.176	−25.0
27	−0.171	−24.7
30	−0.166	−24.2
34	−0.168	−25.8

The most-used BDT features (w_η^{CoG}, w_ϕ^{CoG}, ET1, e_{11}/e_{33}, e_{32}/e_{35}, see `FunWithxgboost/config.yaml` lines 48–73) all shift in the direction that makes converted photons look more background-like. The BDT training is therefore implicitly rejecting converted photons even though no truth conversion information enters the classifier.

7 Physics Interpretation

The four analyses (fractions, response, isolation, shower shape) assemble into a single, self-consistent picture of converted photons in sPHENIX dictated by the ~ 1.4 T solenoid:

1. The conversion probability is flat in p_T and rises with $|\eta|$. The $\sim 17\%$ central-barrel value is consistent with the quoted 10–20% X_0 of integrated material upstream of the EMCal.

2. The magnetic field opens the e^+e^- pair in ϕ (but not in η). One of the two arms has a finite probability of curling outside the cluster footprint, leaving partial energy in the EMCal. This produces a long low-R tail in the energy response — not a shift of the core (the narrow-core Gaussian sits at $\mu = 1.00$ for both categories) — and a bimodal $\Delta\phi$ distribution for the converted category.

3. The lost-arm shower lands inside the $R = 0.3$ isolation cone and inflates `cluster_iso_03_emcal` by ~ 0.5 GeV on average. Meanwhile the HCal sub-components move in the opposite direction (the two overlapping e^+e^- showers are absorbed more compactly in the EMCal), confirming the same spatial leakage pattern. The parametric iso cut therefore rejects converted photons ~ 14.6 pp more often than unconverted ones inclusively, rising to ~ 23 pp at low p_T.

4. The wider, two-lobed transverse shower profile is picked up most cleanly by w_ϕ^{CoG} (mean bimodality, +91% relative shift) and e_{32}/e_{35} (tightens the central 3×2 fraction). The BDT absorbs these features through its training on jet-MC background and produces a well-separated low-score tail whose effect on the tight selection grows with p_T, reaching -26% at 23–34 GeV.

5. All four conversion signatures share a common p_T evolution: the effect is strongest at low p_T (large magnetic opening relative to the cluster/cone scale) and fades at high p_T where the e^+e^- pair is collimated. The BDT shift is the exception — it grows with p_T — because the parametric cut itself becomes tighter at high p_T and the low-score tail of converted photons sits on the wrong side of an increasingly stringent boundary.

8 Implications for the Isolated-Photon Cross-Section

The converted-photon signatures above enter the PPG12 cross-section measurement as an ID+iso efficiency effect:

- The current nominal efficiency $\varepsilon(\text{reco} \to \text{tight} + \text{iso})$ is computed on simulated signal events that *already* contain conversions at their natural $\sim 17\%$ rate. As long as the GEANT material budget matches reality, the effect is correctly absorbed into the nominal efficiency.

- The converted population is simultaneously penalised by the BDT tight cut (lower mean BDT, growing low-score tail) *and* by the iso parametric cut (EMCal pollution from the lost e^+e^- arm). The PPG12 signal efficiency is therefore a product of two partially anticorrelated selection factors, both of which track the converted fraction.

- A mismatch between the simulated and data material budget (e.g. missing support structure, cable run, coolant volume) would produce a *double* shift: the effective converted fraction would move, and both $\varepsilon_{\mathrm{BDT}}$ and $\varepsilon_{\mathrm{iso}}$ would respond.

The conclusion is that the material budget is one of the key cross-section systematics to control. A dedicated material-budget uncertainty — propagated through the converted fraction to both $\varepsilon_{\mathrm{BDT}}$ and $\varepsilon_{\mathrm{iso}}$ — would correctly cover the combined ID+iso efficiency response.

9 Recommendations and Next Steps

1. **Re-run the isolation study with topo-iso branches populated.** The nominal analysis uses `cluster_iso_topo_04` (`use_topo_iso: 2`); the current study used `cluster_iso_03` by necessity. Populating `cluster_iso_topo_03/04` on this sample (re-running `apply_BDT.C` or the slim-tree builder with the topo branches enabled) will allow the absolute pass-fraction numbers to be tied directly to the nominal cut.

2. **Split the tight-selection efficiency by conversion category.** Apply BDT $> 0.80 - 0.015 \cdot E_T$ and the parametric iso cut of Eq. 2 separately to unconverted, converted, and combined populations to obtain $\varepsilon_{\text{unconv}}$, $\varepsilon_{\text{conv}}$, and the natural-fraction-weighted efficiency. The difference between the weighted and the unconverted-only result is the conversion contribution to the nominal efficiency.

3. **Cross-check BDT data/MC agreement at low BDT values.** The converted-photon population produces a distinct low-score BDT tail. The shape of that tail in *data* (which carries no truth conversion flag) is a direct probe of the simulated material budget: if the MC underestimates the material, the observed low-BDT-score rate will be *higher* than predicted.

4. **Independent toy: artificially remove conversions from MC and propagate.** A zero-conversion counterfactual (e.g. selecting only `particle_converted==0`) gives a direct handle on the converted contribution to the nominal efficiency and to the cross-section.

5. **Repeat on photon5 and photon20.** The pT dependence of the converted-efficiency loss should be cross-checked at 1–14 GeV (`photon5`) and 20–40 GeV (`photon20`) to confirm the crossover at $p_T \sim 25$ GeV and the growing BDT shift.

A Slimtree branches used

- `particle_pid`, `particle_Pt`, `particle_Eta`, `particle_Phi`, `particle_E`, `particle_trkid` — truth four-vector and track id (written at `CaloAna24.cc` ∼957–966).
- `particle_converted` — conversion flag (0/1/2; see Sec. 1).
- `particle_truth_iso_03` — truth isolation E_T in $R = 0.3$.
- `CLUSTERINFO_CEMC.cluster_Et`, `cluster_truthtrkID` — reconstructed cluster E_T and truth-track assignment.
- Shower shape: `weta_cogx`, `wphi_cogx`, `et1`–`et4`, `e11/e33`, `e32/e35`, `CNN_prob`, `cluster_prob`, `merged_prob`.
- `CLUSTERINFO_CEMC_NO_SPLIT.bdt_score` — shower-shape BDT output.
- Isolation: `cluster_iso_03`, `cluster_iso_04`, `.._emcal`, `.._hcalin`, `.._hcalout`.

B Figure inventory

All figures live under `reports/converted_photon_study/figures/`. A compact inventory:

- Fractions: `fractions_vs_pt_eta07.pdf`, `fractions_vs_abseta_eta07.pdf`, `fractions_match_eff_eta07` (plus `_eta15` analogues; `fractions_2d_*`; `fractions_multiplicity_*`).
- Response: `response_shape_grid.pdf`, `response_shape_ET_grid.pdf`, `response_mean_vs_pt.pdf`, `response_sigma_vs_pt.pdf`, `response_tailfrac_vs_pt.pdf`, `response_match_eff_vs_pt.pdf`, `response_eta_dep.pdf`, `response_angular_deltas.pdf`.

- Isolation: `isolation_cluster_iso_03.pdf`, `isolation_cluster_iso_03_emcal.pdf` (plus `_hcalin`, `_hcalout`, `_cumulative`; R=0.4 analogues), `isolation_mean_vs_pt_R003.pdf`, `_R004.pdf`, `isolation_pass_fraction_vs_pt.pdf`.
- Shower shape (each variable has `_inclusive`, `_mean` and a 12-panel p_T-binned PDF): `showershape_weta_co` `showershape_wphi_cogx*.pdf`, `showershape_et1*.pdf` to `et4*.pdf`, `showershape_e11_over_e33*.pdf`, `showershape_e32_over_e35*.pdf`, `showershape_et2_over_et1*.pdf` to `et4_over_et1*.pdf`, `showershape_CNN_prob*.pdf`, `showershape_cluster_prob*.pdf`, `showershape_merged_prob*.pdf`, `showershape_bdt_score_nosplit*.pdf`.